

Joins.

Start @ 9:03

Recap:

Subqueries

Group by

if null

not null

Aggregation

Trim

if else.

case when.

Agenda

Having.

Join.

Types of joins

Unions.

Having:

show all customer who made
overall purchase > 20

Users	Purchase
A	1
A	8
A	15
B	23
C	10

A	24
B	23
C	12

X

c	2
---	---

When you want to filter on the aggregation.

Select user
sum(purchase) overall sales

from DB table

group by user

having sum(purchases) > 20.

Order of writing

Select
Distinct
Aggregation
from
where
Group by
Having
order by
limit
offset

Order of exec.

From
where
Group by
Aggregation
Having
Distinct
Select
Order by
offset
limit

Joins:

Cid	Pid	Sales
1	P4	10
2	P4	20
2	P5	5

Cusid	Name
1	A
2	B
3	C
4	D

B overall sales $\rightarrow$ 25.

Types:

Inner

Left

Right

Outer

Inner. : All common rows in both table.

A	Sales	B	Sport
1		2	
2		3	
3		4	
3		5	

A	B
2	2
3	3
3	3
4	4

Cus	Fname	Sales
①	Raj	10
2	Tharish	20
2	Tharish	40
3	Suhoil	8

Cus-id	Lname
1	Kumar
2	Batcha
3	Ahmed
4	Suresh.

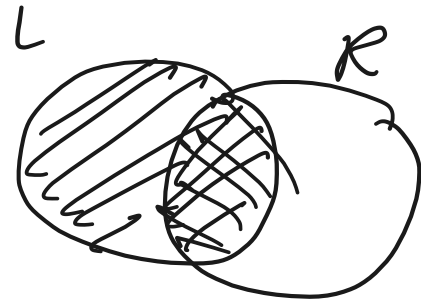
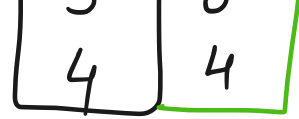
Cus	Fname	Sales	Lname
①	Raj	10	Kumar
2	Tharish	20	Batcha
2	Tharish	4	Batcha
3	Suhoil	8	Ahmed

L. Join: All rows on left table and only matching rows from right

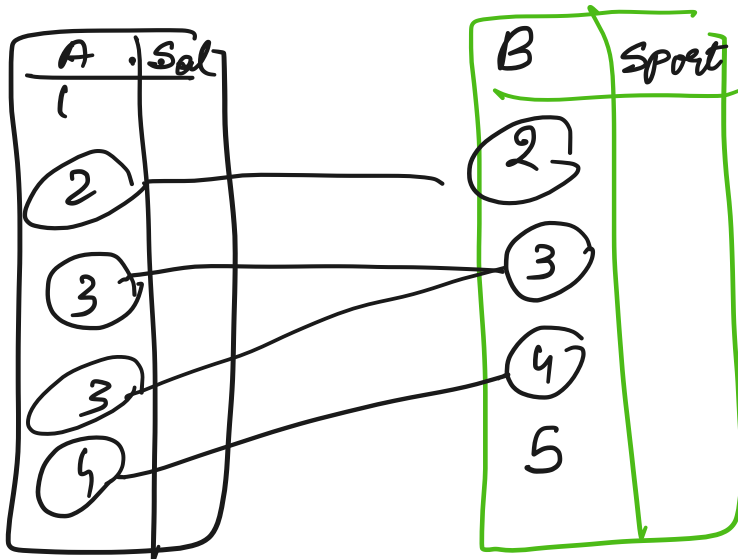
A	Sales
1	
2	
3	
3	

B	Sport
2	
3	
4	

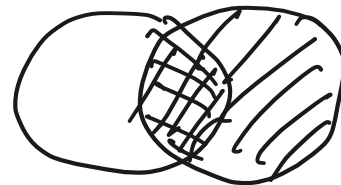
A	B
1	NULL
2	2
3	3
3	3



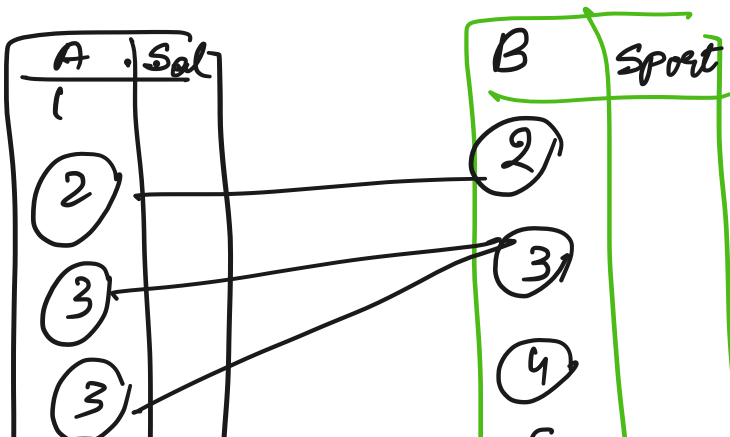
Right Join: Every rows on right table and only matching from left table



A	B
2	2
3	3
3	3
4	4
NULL	5



Outer Join: Every row on left, every row on right and matching.



A	B
1	NULL
2	2
3	3
3	3

→ left

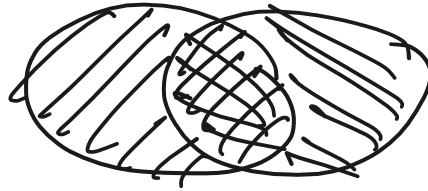
→ inner

4	5
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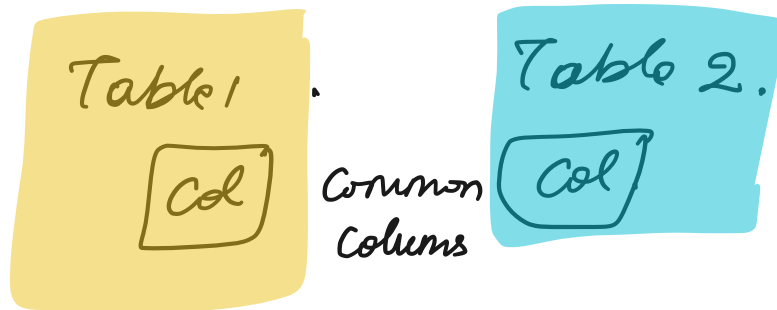
5	
---	--

4	4
NULL	5

→ right



Joins Syntax:



Select

table 1. col 1

table 2. col 2

from

DB. table 1

join

DB. table 2

on

table 1. col = table 2. col



Common column between

2 tables (ideally P.K. & F.K.)

left table

could be any join

right table

Order of writing

Select

Distinct

from

Join

where

group by

Having

Order by

limit

offset

Order of Exec

From, Join

where

Group by

Aggregation

Having

Distinct

Select

Order by

offset

limit

Sales

Tr. id	Cus. id	Sales
T1	C1	10
T2	C1	20
T3	NULL	5
T4	C4	15

Cus. table

Cus. id	Name
C1	John
C3	Mike

Unions:

Cus-id	Sales	Name
10	15	A
11	20	B
12	5	

Cus-id	Sales	Name
10	15	A
11	20	B
15	4	C

Unions.

Cus-id	Sales	Name
10	15	A
11	20	B
12	5	

Cus-id	Sales	Name
10	15	A
11	20	B
15	4	C

Union
distinct

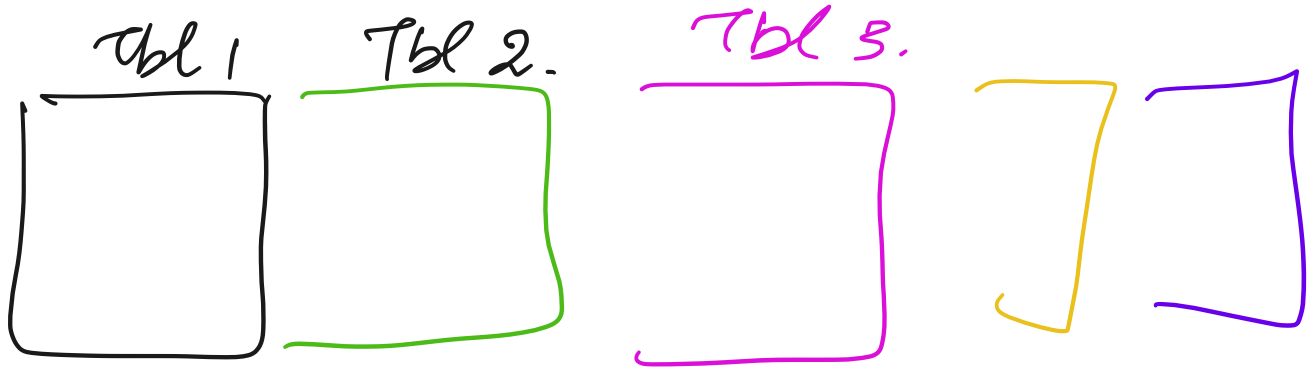
Cus-id	Sales	Name
10	15	A
11	20	B
12	5	
15	4	C

Common
row will
be combined

Union → combine common columns to
make sense

Combine same data type columns

Joining multiple tables



Do all 4 kinds of joins

T1

id	
1	
1	
2	
2	
NULL	

T2

id	
1	
2	
3	
NULL	